

HANDOUT TITLE

SPRING 2026 · COURSE NAME

1 Energy and Conservation

A particle of mass m moves in a one-dimensional potential $V(x)$. The total mechanical energy is

$$E = \frac{1}{2}mv^2 + V(x), \quad (1)$$

which is conserved when no external forces act.

Definition 1.1. A force is *conservative* if it can be written as the negative gradient of a potential, $F(x) = -dV/dx$.

Theorem 1.2. *For a conservative force, the total mechanical energy E is constant along any solution of the equation of motion.*

Remark 1.3. The converse fails in general: dissipative systems admit no such potential, and E is not conserved.

1.1 Potential Energy Diagrams

A plot of $V(x)$ against x summarizes the qualitative motion at a glance: the particle is confined to regions where $E \geq V(x)$, and turning points occur where $E = V(x)$, as illustrated in Figure 1.

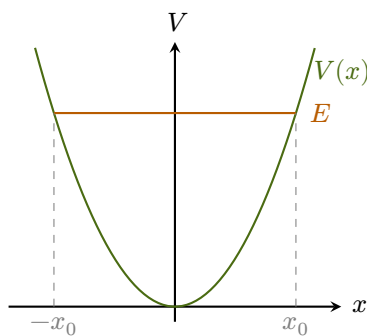


Figure 1. Harmonic potential (olive) with energy level E (orange) and classical turning points $\pm x_0$ (gray).

The subsection heading above sits one size step below the section title, while remaining at body size — enough to mark a finer level when the handout is used for lecture notes.

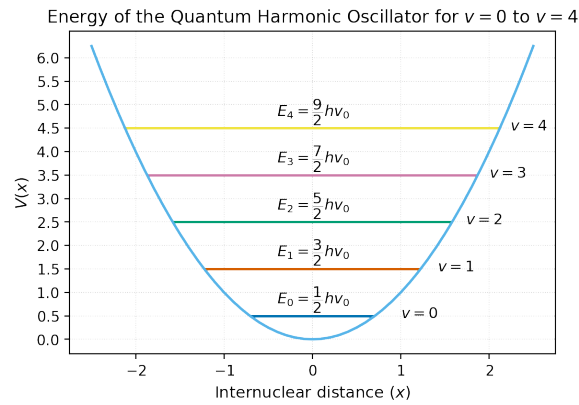


Figure 2. Schematic figure for the topic.

Exercise 1.1 - Simple harmonic oscillator

Consider a particle of mass m in the potential $V(x) = \frac{1}{2} k x^2$.

- Write down the equation of motion using Newton's second law.
- Show that $E = \frac{1}{2} m v^2 + \frac{1}{2} k x^2$ is conserved along solutions.

The result can be verified by differentiating E with respect to time and using the equation of motion from part (a).

- Find the angular frequency of small oscillations.

Exercise 1.2 - Inclined plane

A block of mass m slides down a frictionless inclined plane that makes an angle θ with the horizontal.

- Find the acceleration along the plane.
- How long does it take to slide a distance L from rest?
- What is the speed at the bottom?

2 Circular Motion

In uniform circular motion, the centripetal acceleration has magnitude $a_c = v^2/r$, directed toward the center of the circle.

Exercise 2.1 - Circular orbit

A satellite moves in a circular orbit at radius r around a planet of mass M .

- a) Show that the orbital speed is $v = \sqrt{GM/r}$, where G is Newton's gravitational constant.
- b) Express the orbital period T in terms of r , M , and G .

3 Numerical Example

The `python` environment typesets code with Gruvbox Light syntax highlighting. The short script below verifies the orbital speed and period from Exercise 3 numerically for three representative radii.

```
1 import numpy as np
2
3 G = 6.674e-11      # gravitational constant (m^3 kg^-1 s^-2)
4 M = 5.972e24      # Earth mass (kg)
5
6 orbits = {"Low orbit": 6.40e6,
7           "ISS (approx)": 6.78e6,
8           "GEO": 4.22e7}
9
10 for name, r in orbits.items():
11     v = np.sqrt(G * M / r)          # orbital speed [m/s]
12     T = 2 * np.pi * r / v / 3600  # orbital period [hours]
13     print(f"{name:14s} v = {v/1e3:.3f} km/s T = {T:.2f} h")
```